

Memory

Memory

- **Definition:**

Memory is often defined as the application of learning over time. It refers to the ability to **encode, store, and retrieve information** when needed.

- If memory was nonexistent, everyone would be a stranger to you; every language foreign; every task new; and even you yourself would be a stranger.

Key Questions about Memory:

How does memory work?

Memory works through three main processes:

- *Encoding*: Converting information into a form the brain can process.
- *Storage*: Retaining that information over short or long periods.
- *Retrieval*: Accessing the stored information when needed.

How much do we remember?

- Our memory is selective. We tend to remember information that is meaningful, emotional, or frequently rehearsed.
- Short-term memory is limited (about 7 ± 2 items), while long-term memory has a much larger capacity.

Key Questions about Memory:

How can we recall more?

- Use **retrieval cues** (like associations, context, or mnemonics).
- Practice **repetition and rehearsal**.
- Organize information into **chunks** or meaningful categories.

How can we better remember tasks in the future? (*Prospective memory*)

- Use reminders or external aids (calendars, alarms, sticky notes).
- Form habits by linking tasks with routine cues.
- Visualize yourself completing the task (“mental rehearsal”).
- Use the “**implementation intention**” technique (e.g., “If it’s 9 a.m., then I will take my medicine”).

Multiple Systems Model of Memory

Declarative Memory (Explicit Memory)

- **Definition:** Memory that can be consciously recalled and verbalized.
- **Brain areas:** Depends heavily on the **medial temporal lobe** and **hippocampus**.
- **Types:**
 - **Episodic Memory** → Memory of personal experiences and events tied to a specific time and place. (Remembering your 10th birthday party, your first day at school, or a trip you took.).
 - **Semantic Memory** → Memory of facts, meanings, concepts, and general knowledge about the world. (e.g., *Paris is the capital of France*).

Feature	Episodic Memory	Semantic Memory
Content	Events & personal experiences	Facts, concepts, general knowledge
Time & Place	Specific (when/where it happened)	Not linked to time or place
Example	“I remember my graduation day.”	“A graduation is a ceremony for students.”
Brain region	Hippocampus, medial temporal lobe	Temporal lobe, neocortex
Conscious recall	Mental “re-experiencing” (autobiographical)	Just knowing (factual recall)

- Declarative memory or explicit memory is a memory system that is controlled consciously, intentionally, and flexibly.
- Declarative memory generally involves some effort and intention, and we can employ memory strategies such as mnemonics to recall information.
- Examples include recalling the name of an old friend, remembering a list of items to pick up at the store, remembering information for a test, learning a phone number, and recalling your ATM password

2. Non-Declarative Memory (Implicit Memory)

- **Definition:** Memory expressed through performance, not conscious recollection.
- **Brain areas:** Involves **cerebellum, basal ganglia, amygdala**, and cortical areas.
- **Types:**
 - **Procedural Memory** → Skills and habits (e.g., *riding a bicycle, typing*).
 - **Classical Conditioning** → Learned associations between stimuli (e.g., *Pavlov's dog salivating to a bell*).
 - **Priming** → Exposure to one stimulus influences response to another (e.g., *seeing "doctor" makes it easier to recognize "nurse"*).
 - **Non-associative Learning** → Habituation (decreased response) and sensitization (increased response).

- Nondeclarative memory or implicit memory is a memory system that influences **our current perceptions and behavior without our knowledge, awareness, or intention.**
- Nondeclarative memory is not used intentionally and involves no effort. It is assessed with an implicit memory test in which the individual is unaware she or he is taking a memory test.
- Examples of nondeclarative memory include riding a bike.

Procedural Memory - “Knowing how” (skills and actions).

- **Definition:** Memory for **skills, habits, and processes** involved in completing a task.
- **Nature:** Develops gradually with practice → becomes **automatic** and doesn't require conscious thought.

Examples:

- Playing the piano 
- Typing on a keyboard 
- Riding a bicycle 
- Hitting a tennis ball 

Brain regions: Mainly the **basal ganglia** and **cerebellum**.

Classical Conditioning - “Associations” (stimulus → response learning).

- **Definition:** Memory for **associations formed between two stimuli**, so that one stimulus predicts another.
- **Nature:** An initially neutral stimulus comes to trigger a response after being paired with a meaningful stimulus.

Examples:

- Pavlov’s dogs 🐕 → Bell (neutral stimulus) + Food (unconditioned stimulus) → Salivation (conditioned response).
- Feeling anxious when you hear a dentist’s drill, because it’s been paired with past pain.

Brain regions: Involves the **amygdala** (emotional conditioning) and **cerebellum** (motor conditioning).

Priming - unconscious influence of prior exposure.

- **Definition:**

An **automatic, unconscious process** where prior exposure to a stimulus influences the response to a later stimulus.

- **Key Features:**

- Happens **without deliberate intention**.
- Enhances **speed, accuracy, and fluency** of responses.
- Often measured in tasks like word recognition, picture identification, or reaction time.

Types of Priming

- **Repetition Priming**

- Exposure to a stimulus makes it easier to process again.
- **Example:** You read the word “*pretzel*” once, and the next time you see it, you read it aloud **faster**.

- **Semantic Priming**

- Response to a stimulus is faster when it is meaningfully related to a previously seen stimulus.
- **Example:** Seeing the word “*doctor*” makes you quicker to recognize the word “*nurse*”.

- **Perceptual Priming** (extra type worth noting)

- Recognition of a stimulus is improved when its **physical form** is repeated.
- **Example:** Recognizing a blurred picture of an object faster if you’ve seen the clear version before.

Non-associative Learning

1. Habituation

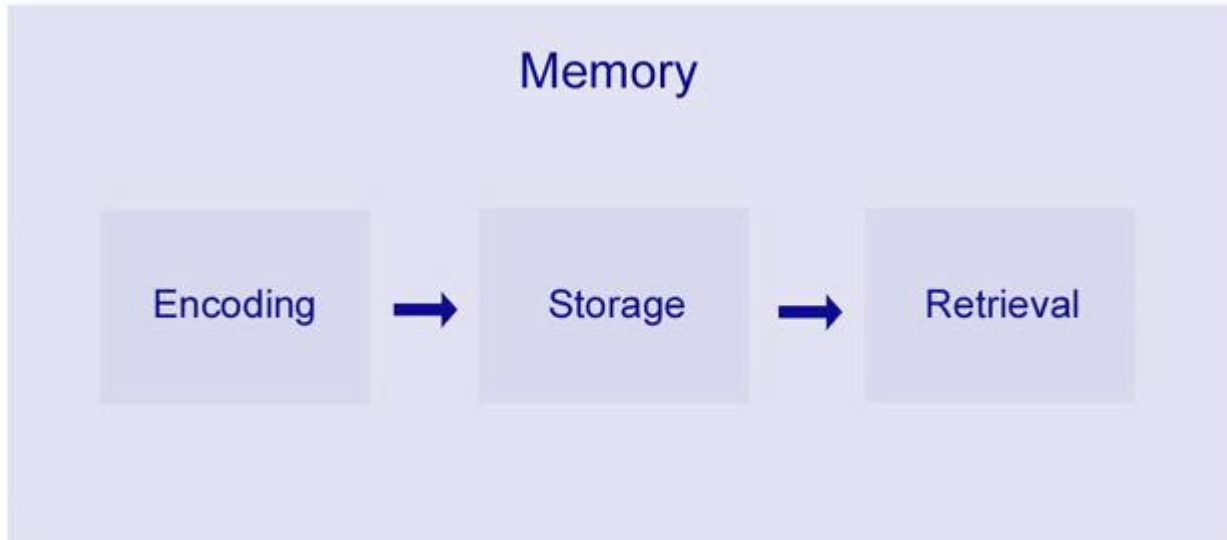
- **Definition:** A **decrease in response** after repeated exposure to the same stimulus.
- **Purpose:** Helps us ignore irrelevant or non-threatening stimuli.
- **Example:**
 - You stop noticing the ticking sound of a clock after sitting in the room for a while.
 - A person living near a train track stops reacting to the train noise.

2. Sensitization

- **Definition:** An **increase in response** after repeated or particularly strong exposure to a stimulus.
- **Purpose:** Heightens alertness to potentially dangerous or important stimuli.
- **Example:**
 - After being shocked once, you become **more responsive** (even jumpy) to small noises.
 - If someone scratches a chalkboard, the irritation becomes **worse** the more it's repeated.

Declarative (Explicit)	Non-Declarative (Implicit)
Conscious, intentional recall	Unconscious, automatic recall
Facts & events	Skills, habits, priming, conditioning
Depends on hippocampus & medial temporal lobe	Depends on cerebellum, basal ganglia, amygdala
Example: Remembering a lecture	Example: Knowing how to play piano

Information Processing Models



Keyboard
(Encoding)



Disk
(Storage)



Monitor
(Retrieval)

Sequential Process



Encoding

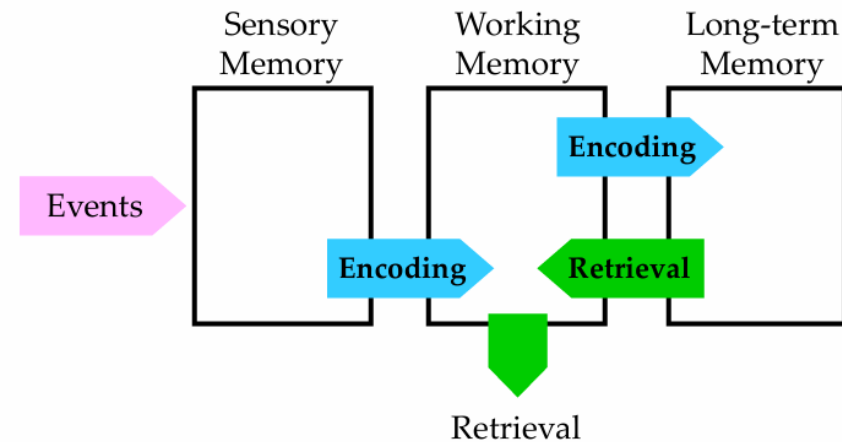
- Encoding is the process by which we place the things that we experience into memory.
- Unless information is encoded, it cannot be remembered.
- If you have encoded someone's name in the category of “friends in my class,” you might not be able to identify that person when you see them in the Lulu Mall.

Storage

- It is the process of holding information in memory to be processed or used.
- Three-stage of memory includes
 - a) sensory memory,
 - b) short-term memory/working memory,
 - c) long-term memory.

Storage: Retaining Information

- Storage is at the heart of memory. Three stores of memory are shown below:



- Some memories we will hold for years, other memories we hold only long enough to use the information, such as looking up a phone number and retaining it long enough to place the call.

Sensory memory

- The **briefest form of memory**, lasting only **a fraction of a second to a few seconds**.
- It holds sensory information (from vision, hearing, touch, etc.) just long enough for the brain to **process and decide** whether to transfer it to **short-term memory**.

Functions of Sensory Memory

- Gives the brain time to **process incoming sensations**.
- Allows us to perceive the world as a **continuous, unbroken stream of events** (instead of disjointed flashes).
- Acts as a **buffer** between raw sensory input and conscious awareness.

Sensory Memories

The duration of sensory memory varies for the different senses.

- Iconic memory for visual information
- Echoic memory for auditory information
- Haptic for sense information

Iconic
0.5 sec. long

Echoic
3-4 sec. long

Haptic
< 1 sec. long



- **Iconic Memory (Visual):**
 - Holds visual information for about **0.5 seconds**.
 - Example: Seeing a lightning flash linger in your vision for a split second.
 - **Echoic Memory (Auditory):**
 - Holds auditory information for about **3–4 seconds**.
 - Example: When you say “What?” but then realize you actually heard what was said.
 - **Haptic Memory (Touch):**
 - Holds tactile information for about **1–2 seconds**.
 - Example: Still feeling the texture of an object right after releasing it.
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- **Sensory memory** is like the brain’s “**raw data buffer.**”
 - If attended → information moves to **Short-Term Memory (STM)**.
 - If ignored → it quickly disappears.

W T E S

Q A H M

O P W C

- You were asked to report either the entire matrix (whole report) or cued with a tone to report one random line (partial report).
- When required to report one random line, participants could report nearly all the items in that line, regardless of which line was selected, suggesting that the capacity of the visual sensory register is large but remains present for only a short duration.
- Information in the sensory register gradually decays over time

Short-Term Memory (STM) & Working Memory (WM)

From Sensory Memory to STM

- Most sensory information fades quickly.
- Only the information we **pay attention to** moves into **short-term memory**.

Short-Term Memory (STM)

- **Definition:** A temporary storage system where **small amounts of information** are kept for **seconds to under a minute**.
- **Duration:** 15–30 seconds (unless rehearsed).
- **Capacity:** Limited ($\sim 7 \pm 2$ items).
- **Nature:** Information is not permanently stored → it's available only for immediate use.
- **Example:** Remembering a phone number just long enough to dial it.

Working Memory (WM)

- **Definition:** The set of **processes applied to information in STM** to make sense of it, modify it, interpret it, and prepare it for long-term storage.
- **Function:** Not just holding, but **actively manipulating** information.
- **Examples:**
 - Doing mental arithmetic (23 + 49).
 - Understanding a sentence as you read.
 - Following directions while driving.

- **STM** is like a **notepad** where information is temporarily written.
- **WM** is like a **notepad + workspace**, where you can **write, erase, rearrange, and connect** information.

Relationship Between STM & WM

- **STM = storage** (passive holding of information).
- **WM = storage + processing** (active manipulation of information).

The “Magic Number” – George Miller (1956)

- **Finding:**

George Miller proposed that the capacity of **short-term memory** is about **7 ± 2 items** (between 5 and 9 items) for most adults.

- **Meaning:**

Humans can only hold a limited number of separate pieces of information in **short-term memory** at one time.

Chunking

- **Definition:**
The process of **grouping individual pieces of information into larger, meaningful units (chunks)** so they can be remembered more easily.
- **Why It Works:**
Instead of storing each small item separately, the brain stores a **chunk** as one unit, effectively expanding STM capacity.
- **Examples:**
 - Remembering digits **1-9-4-7-1-9-4-8** as **1947** and **1948** (two chunks instead of eight items).
 - Remembering letters **F-B-I-C-I-A-U-S-A** as **FBI – CIA – USA**.
- **Effect:**
Chunking allows for **easier encoding, storage, and retrieval** in both STM and transfer to LTM.

How could you chunk these examples?

- 1 8 0 0 2 2 5 5 2 8 8
- 1-8 0 0- 2 2 5- 5 2 8 8
- C B S I R S M T V P B S D M V
- C B S I R S M T V P B S D M V
- 1-4-9-2-1-7-7-6-1-9-9-9-2-0-0-5
- 1492-1776-1999-2005

How long can this information stay in STM?

Why Memories Disappear

- Information in **STM** is **fragile** → unless it's **rehearsed** or **made meaningful**, it fades quickly.
- Memories are more likely to last if:
 - You **continually rehearse them** (repetition).
 - They are **meaningful** or emotionally important, so they are encoded deeply and stored quickly in **LTM**.

Rehearsal

- **Definition:**
The process of **repetitively verbalizing or mentally thinking about information** to keep it active in STM and help encode it into LTM.
- **Types of Rehearsal:**
 - **Maintenance Rehearsal**
 - Simple repetition of information.
 - Keeps information in STM but doesn't guarantee transfer to LTM.
 - Example: Repeating a phone number until you dial it.
 - **Elaborative Rehearsal**
 - Linking new information to existing knowledge or making it meaningful.
 - Leads to deeper encoding → better transfer to LTM.
 - Example: To remember the word "*photosynthesis*", connecting it with "plants making food with sunlight."

Long-term Memory

- Memory system that stores information for **extended periods of time**—from days to years, and often a lifetime.
- Information that makes it past **Short-Term Memory (STM)** through rehearsal or meaningful encoding can enter LTM.

Key Features

- **Duration:** Very long, potentially permanent.
- **Capacity:** Extremely large; no known upper limit to how much we can store.
- **Forgetting:**
 - Some information may be forgotten due to decay, interference, or retrieval failure.
 - Other memories (especially meaningful, emotional, or well-practiced ones) may remain accessible for a lifetime.

Examples

- Remembering your **childhood home**.
- Knowing how to **ride a bicycle** years after learning.
- Retaining **facts, skills, and life experiences** for decades.

Some memories are deeply encoded and stay with us permanently.

Retrieval

- Retrieval = bringing stored info back into awareness.
- Influenced by context, state, rehearsal, order, and interference.
- Neural basis = **LTP**, where repeated activation strengthens synapses.
- Multiple brain regions work together

- Retrieval is the process of **reactivating and accessing information** that has been stored in memory.
- Without retrieval, memory storage would be useless because we couldn't bring past experiences or knowledge back into awareness.

Factors Influencing Retrieval

- **Context** → We remember better when we are in the same environment where the memory was encoded. (*Context-dependent memory*).
- **Physiological State or Mood** → Recall improves when we are in the same emotional or physical state as during learning. (*State-dependent memory, mood-congruent memory*).
- **Order of Presentation** →
 - **Primacy effect**: Better recall of items at the beginning of a list.
 - **Recency effect**: Better recall of items at the end of a list.
- **Rehearsal** → More rehearsal strengthens memory traces.
- **Interference** → Competing information can block retrieval:
 - *Proactive interference*: Old information disrupts recall of new.
 - *Retroactive interference*: New information disrupts recall of old.

Biological Basis of Memory

- **Memories are stored in synaptic connections** between neurons.
- **Long-Term Potentiation (LTP):**
 - Refers to the **strengthening of synaptic connections** between neurons after frequent stimulation.
 - “Neurons that fire together, wire together.”
 - Makes future communication across the synapse more efficient → the biological foundation of learning and memory.

Feature	Short-Term Memory (STM)	Long-Term Memory (LTM)
Duration	15–30 seconds	Lifetime (potentially)
Capacity	7 ± 2 items	Unlimited
Encoding	Acoustic/visual	Semantic (meaning-based)
Conscious use	Active, immediate tasks	Knowledge, skills, past experiences
Example	Remembering a number temporarily	Remembering your graduation day